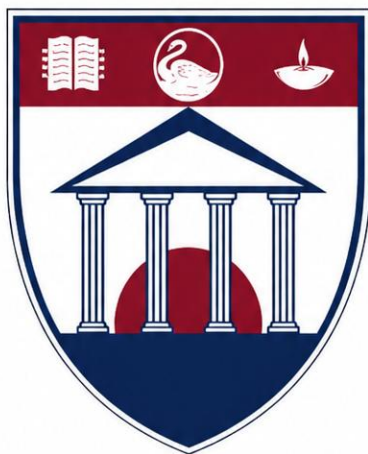


REPORT OF Cybersecurity Internship: Development of Security-Based Applications

Submitted To:

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfillment of the requirement of degree of B.Tech CSE



BY:

Roll Number of Student : 25scs1003001109

Name of Student : YASH KUMAR SHARMA

Batch : 2025-2029

CANDIDATE'S DECLARATION

I, **YASH KUMAR SHARMA**, do hereby declare that the internship report titled "Development of Cybersecurity Tools and Secure Applications" has been completed by me to fulfill the requirement for the award of the degree of Bachelor of Technology in CSE with Cyber Security. The work presented in this report is based on the cybersecurity internship tasks assigned by Thiranex and the project requirements provided during the internship.

I affirm that this report has been prepared for academic submission and that I shall be responsible for the authenticity and proper acknowledgement of the material included in it.

Signature: _____

Student Name: YASH KUMAR SHARMA

Roll No: 25SCS1003001109

ACKNOWLEDGEMENT

I express my sincere gratitude to everyone who supported me during my cybersecurity internship. I am thankful for the guidance, feedback, and practical learning opportunities that helped me understand cybersecurity concepts and their implementation through project-based work.

I also express my gratitude to my parents, teachers, and the internship organization for their encouragement and support throughout this learning experience.

Signature: _____

Student Name: YASH KUMAR SHARMA

Roll No: 25SCS1003001109

4. PROJECT DETAILS

4.1 Introduction

The cybersecurity internship provided practical exposure to different areas of cybersecurity and secure software development. Four project areas were covered: Password Strength Analyzer, Vulnerability Scanner, Phishing Email Detection Model, and Secure Login System. Together, these projects introduce password security, network scanning, phishing detection, machine learning, secure authentication, password hashing, SQL injection prevention, and session management.

4.2 Organization Profile

Organization Name: Thiranex

Thiranex provides internship opportunities that allow students to gain practical experience through project-based learning. The internship focuses on developing technical knowledge and practical skills by working on real-world-oriented projects and cybersecurity tasks.

4.3 Problem Statement

Cybersecurity threats such as weak passwords, exposed network services, phishing emails, and insecure authentication can lead to unauthorized access and data loss. The internship projects address these common security concerns by building simple tools and applications that demonstrate preventive and analytical security techniques.

The project set focuses on:

- Analyzing the strength and complexity of passwords.
- Identifying open ports and basic network security risks.
- Detecting phishing emails using machine learning and email features.
- Developing a secure authentication system with password hashing and session management.

4.4 Project Objectives

- Understand password security and password strength analysis.
- Learn basic network and port scanning concepts.
- Identify potential vulnerabilities associated with network services.
- Understand phishing attacks and email classification.
- Apply machine learning concepts to a cybersecurity problem.
- Develop a secure login and registration workflow.
- Implement secure password hashing using bcrypt or an equivalent secure approach.
- Understand input validation and SQL injection prevention.
- Implement session management and logout functionality.

4.5 Scope of the Project

The scope of the internship project is focused on the development and implementation of basic cybersecurity tools and secure applications. The work covers four complementary areas: password security, network vulnerability assessment, phishing detection, and secure user authentication.

The projects are intended to provide practical understanding of common cybersecurity problems and demonstrate how programming can be used to develop simple solutions for security analysis and protection.

1. Password Strength Analyzer

This project evaluates the strength of user-entered passwords using security-related characteristics. The task specification emphasizes password length, complexity, uniqueness, and stronger alternatives.

Key Features:

- Check password length, complexity, and uniqueness.
- Evaluate the presence of uppercase letters, lowercase letters, numbers, and special characters.
- Classify passwords into understandable strength levels such as Weak, Medium, or Strong.
- Suggest stronger password alternatives.
- Optionally support a database to help prevent reuse of old passwords.

Expected Outcome:

The project helps users understand password security and basic cryptography concepts and encourages the creation of stronger passwords.

2. Vulnerability Scanner (Mini Project)

This project builds a simple script to detect common vulnerabilities or weaknesses in a web application or network environment. The task specification focuses on basic scanning and reporting.

Key Features:

- Scan for open ports or weak configurations.
- Identify outdated software versions where such information is available.
- Generate a simple vulnerability report.
- Develop practical understanding of penetration testing and vulnerability assessment concepts.

Expected Outcome:

The project provides introductory experience with network security, port scanning, security warnings, and vulnerability assessment.

4.5 Project Details (Continued)

3. Phishing Email Detection Model

This project develops a machine-learning-based classifier using Scikit-learn to distinguish suspicious phishing emails from legitimate emails.

Key Features:

- Train on a dataset containing phishing and legitimate emails.
- Extract and analyze email features such as URLs and keywords.
- Classify emails as “Phishing” or “Safe”.
- Display model accuracy and a confusion matrix.
- Use textual content and URL-related features to support classification.

Expected Outcome:

The model demonstrates how machine learning can be applied to cybersecurity for phishing detection. Performance can be evaluated using accuracy and a confusion matrix.

4. Secure Login System

This project develops a secure login web application designed to protect user accounts through secure authentication practices.

Key Features:

- User registration and login using securely hashed passwords (bcrypt or Argon2).
- Basic input validation.
- Protection against SQL injection through safe database interaction.
- Session management with logout functionality.
- Optional Two-Factor Authentication (2FA).

Expected Outcome:

The secure login system demonstrates secure authentication, password hashing, input validation, session management, and protection against common attacks, thereby reducing unauthorized access risks.

Project Task Evidence

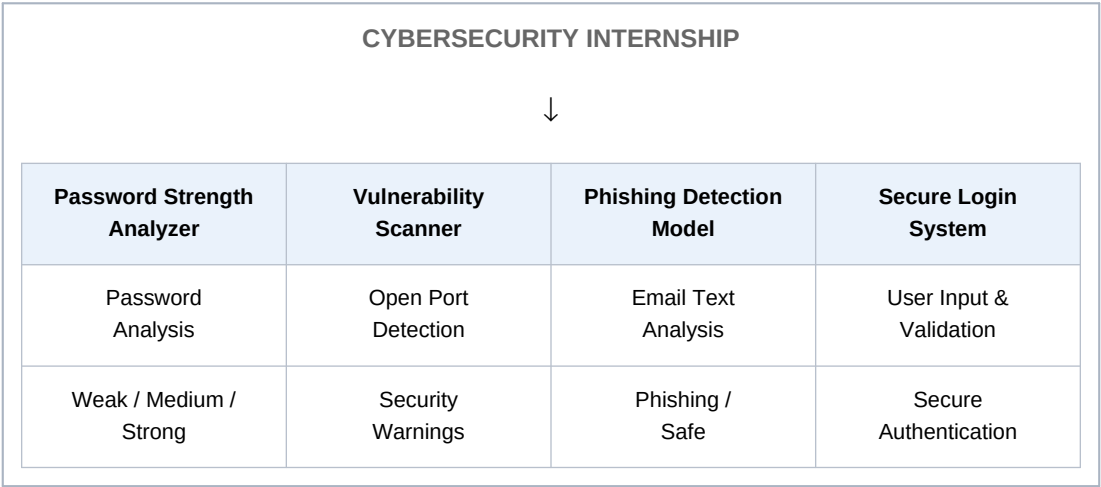
The report is based on the four internship task specifications supplied for this submission. The supplied task screenshots cover the Password Strength Analyzer, Vulnerability Scanner, Phishing Email Detection Model, and Secure Login System requirements.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Main programming language for cybersecurity tasks
Regular Expressions (re)	Password pattern and complexity checking
Socket Module	Network communication and port scanning
Pandas	Data handling for machine-learning workflows
Scikit-learn	Machine learning and email classification
Flask	Web application development
SQLite	Database management
bcrypt	Secure password hashing
Visual Studio Code	Code development and testing
GitHub	Project hosting and version control

4.7 System Architecture

The internship work consists of four independent cybersecurity projects connected by a common learning objective: applying programming to practical security problems.



4.8 Methodology

Step 1: Requirement Analysis

The requirements of each cybersecurity task were studied and the required features were identified.

Step 2: Design

A simple design was prepared for each project based on its objective and expected outcome.

Step 3: Development

The projects were developed using Python and appropriate libraries. Regular expressions support password analysis; socket-based logic supports port scanning; machine-learning libraries support email classification; and Flask, SQLite, and bcrypt support secure authentication.

Step 4: Testing

Each project was tested with sample inputs to verify that its intended functionality was working.

Step 5: Result Analysis

The outputs were reviewed to determine whether the expected functionality and security-oriented behavior were achieved.

4.9 Expected Outcomes

Project	Expected Outcome
Password Strength Analyzer	Analyzes passwords and classifies them by strength based on security requirements.
Vulnerability Scanner	Identifies open ports and provides basic security warnings about potentially insecure services.
Phishing Email Detection	Classifies emails as Phishing or Safe based on textual content and suspicious patterns; also provides user alerts.
Secure Login System	Provides password hashing, input validation, SQL injection protection, session management, and account lockout.

4.10 GitHub Repository

The project submission repository provided for Yash Kumar Sharma is:

https://github.com/yash25scs1003001109-max/2025-2029_yash_1109_3rdSem_CSE20/blob/main/problem4.txt

4.11 Conclusion

The cybersecurity internship provided practical exposure to security-oriented programming and application development. The four projects collectively covered password security, network vulnerability assessment, phishing detection, and secure authentication. The work helped connect programming concepts with common cybersecurity challenges and introduced practical use of tools such as Python, sockets, Scikit-learn, Flask, SQLite, bcrypt, and GitHub.

The internship also strengthened understanding of secure development practices such as password hashing, input validation, SQL injection prevention, session management, and security-oriented testing.

5. BIBLIOGRAPHY / REFERENCES

- Python Software Foundation. Python Documentation.
- Scikit-learn Documentation. Machine Learning in Python.
- Flask Documentation. Flask Web Framework.
- bcrypt Documentation. Password Hashing Library.
- SQLite Documentation. SQLite Database System.
- GitHub. Project Repository and Version Control Platform.

Note: No internship offer letter or completion certificate for Yash Kumar Sharma was supplied with this request, so none has been fabricated or inserted into the report. They can be appended later as supporting documents.